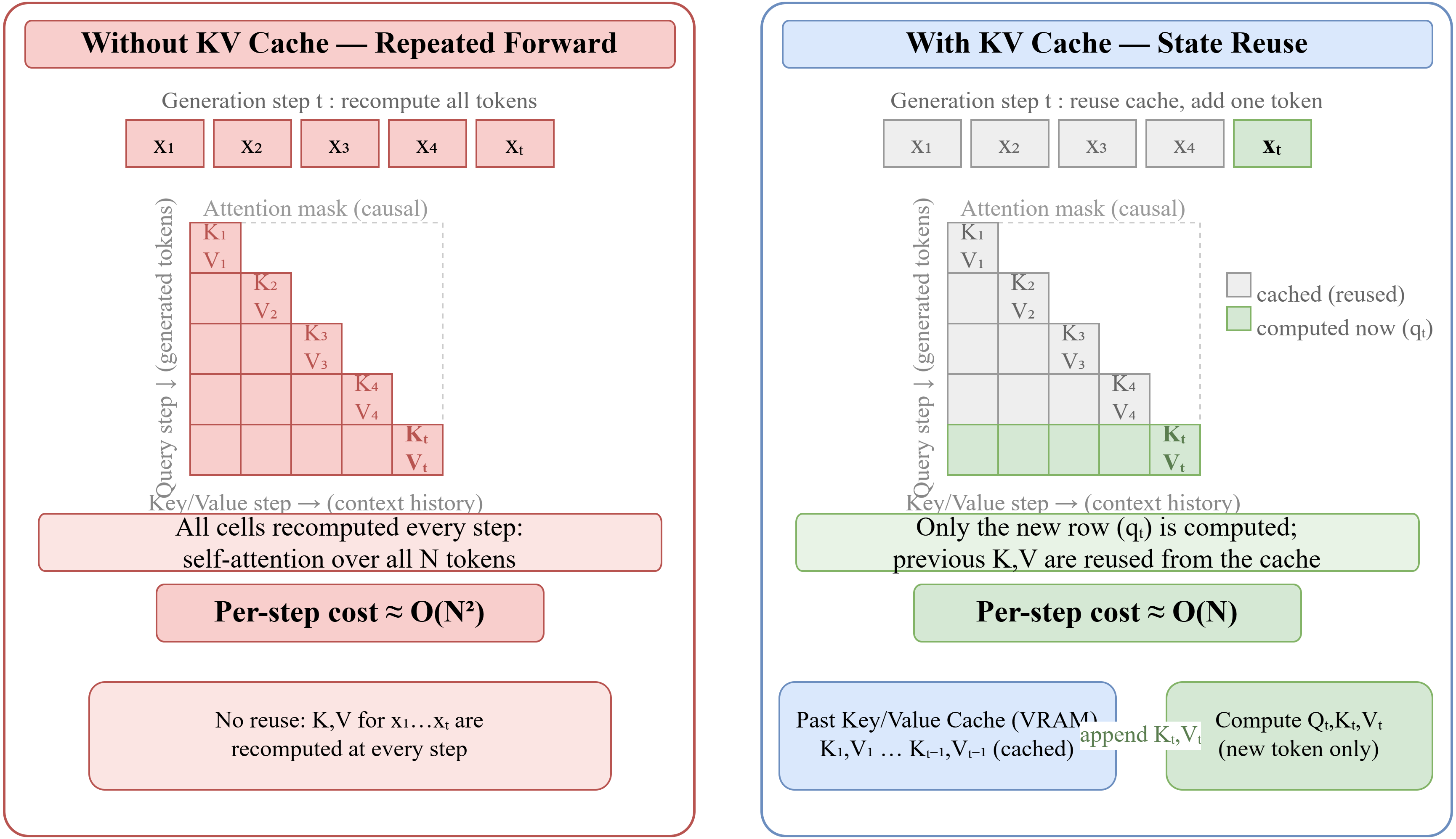

KV Cache reuses cached Key/Value states to cut the per-step decoding cost from $O(N^2)$ to $O(N)$; it does not remove the $O(M^2)$ cost of the initial prompt prefill.